

Identifying Productive Communication with the Public for Scientific Communities in a Fraught Fiscal Time

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This article proposes immediate and long-term solutions to the scientific community's desperate need to better communicate the value of publicly-funded science. Recently, drastic funding cuts to scientific research have resulted in both immediate and future harm to the United States of America. Despite the incredible advances to society made possible by publicly-funded science, scientific communities struggle to explain our value, whether to elected officials, specific communities, or individual people. This article reports results from a workshop conducted at – but not officially sanctioned by - ACM CHI 2025 (Yokohama, Japan). While the urgent situation in the USA spurred this workshop, about half of the workshop participants were from other countries: the inherent riskiness of good science and its ability to challenge the status quo, and the need to publicly communicate the value of our work, are shared struggles around the globe. To that end, the recommendations herein, while immediately applicable to the USA, are intended for broader use as fits each country, community, and communication needs. Much of this is geared specifically towards what scientific communities can accomplish and how they organize. While we can individually accomplish some of these actions, our organizations need to coordinate at a broader scope, both within nations and across the world. We call on national and international scientific organizational leadership to coordinate these opportunities, to connect with other communities currently being punished alongside us, and to develop a coherent support and development system to which individual scientists can contribute.

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1 Introduction

This article proposes immediate and long-term solutions to the scientific community's desperate need to better communicate the value of publicly-funded science. Recently, drastic funding cuts to scientific research have resulted in both immediate and future harm to the United States of America.

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Despite the incredible advances to society made possible by publicly-funded science, scientific communities struggle to explain our value, whether to elected officials, specific communities, or individual people. This article reports results from a ‘Rogue Workshop’ conducted at – but not officially sanctioned by – ACM CHI 2025 (Yokohama, Japan). While the urgent situation in the USA spurred this workshop, about half of the workshop participants were from other countries: the inherent riskiness of good science and its ability to challenge the status quo, and the need to publicly communicate the value of our work, are shared struggles around the globe. To that end, the recommendations herein, while immediately applicable to the USA, are intended for broader use as fits each country, community, and communication needs.

The workshop goal was in part to identify novel strategies for communicating the immense value of publicly-funded science with a diverse array of beneficiaries. These beneficiaries include researchers, designers, practitioners, and industry leaders, as well as the general public who have directly benefited from technology transfer to private industry. Here we focus on recommendations for communicating with the public about the positive impact of our work on their everyday lives.

2 Communicating with the Public – Immediate Actions

The varied beneficiaries of publicly-funded science require that we develop and use different communication and outreach strategies. To that end, and to maximize the overall impact, the following two proposals are designed to reach across different audiences. Our two-pronged approach includes (1) ways to increase the visibility of publicly-funded science in the day-to-day lives of the general population; and (2) actions that will better convey and amplify the harms incurred through haphazard, arbitrary, indiscriminate termination of indispensable resources. The first prong is a longer-term approach to fundamentally changing how publicly-funded science is perceived and represented; the second includes more urgent and immediately actionable suggestions.

2.1 Communicating the ongoing positive impact of publicly-funded scientific work.

Every resident of the USA benefits daily from technological, scientific, and industrial advances that are rooted in or heavily influenced by publicly-funded science. However, we lack a “Made by Publicly-Funded Science” brand, letting our considerable contributions sit quietly behind the company brand names that made our innovations available for purchase and use. As a result, the typical person in the US is unaware of the daily benefits they experience that resulted from the very same scientific communities being actively unfunded. This section focuses potential fixes.

First, we spotlight four excellent examples that can be learned from or adapted. For federal and policy level communications, we recommend examining the Computing Research Association’s “Deconstructing the iPad” workshop for Congress¹ and National Academies of Sciences’ “tire tracks” visualization². Such approaches make the fiscal and social benefits of publicly-funded science concrete and more easily defensible, using the well-tested tools and strategies such organizations have already established. We recommend that scientific organizations consider developing partner materials for scientists to use at regional and state levels as well, to support state policy and legal education efforts. As a more ‘local’ approach, Dr. Kate Starbird’s recent teach-in is an excellent example of how to document the past, present, and future impacts of publicly-funded science³. Similarly, a grassroots movement encourages scientists to write opinion pieces for the communities they grew up in, to spread the word beyond the often large cities that scientific labs are located in⁴. To both those ends, we recommend offering teach-ins in community spaces to increase public

¹<https://cra.org/govaffairs/blog/2011/09/deconstructing-the-ipad/>

²See *Information Technology Innovation: Resurgence, Confluence, and Continuing Impact*, doi:10.17226/25961

³<https://www.hcde.washington.edu/news/article/2025-06-23/foundations-futures-hci-teach-professor-kate-starbird>

⁴<https://www.science.org/content/article/u-s-researchers-are-speaking-science-local-newspapers>

awareness of the valuable work we do, and writing about the impact of these cuts to the communities we were raised in.

Second, public communication efforts are desperately needed, using new approaches to better share the widespread benefits of publicly-funded science. These approaches are necessarily *new* because there has been no similarly coherent, country-wide need for communicating the value of publicly-funded science in the USA. Patchwork efforts to inform the public across various places and times are insufficient, and scientific communities are newly united in their need for clearer communication with the public. As a result, while we recommend some directions below, none of these are specifically limited to a single scientific field; instead, we consider this to be an opportunity to share strategies, approaches, and indeed a ‘brand’ across the various affected sciences.

2.1.1 Example 1 – leveraging striking, contextualized images and videos to illustrate everyday benefits of publicly-funded science. Imagine watching a video of your daily commute on the subway, and suddenly, common objects begin fading out: iPhones disappear, leaving commuters staring at their hands, headphone wires dangling. A baby’s diaper disappears, and liquid starts leaking through their pants onto the floor. Someone breaks out in measles and collapses. The background heartbeat you didn’t realize you were hearing suddenly stops, and a man in a suit suddenly clutches where his pacemaker used to be. The subway car slows down with grinding noises, sparks flashing outside the windows, and finally stops. Lights flicker, and go out. Text appears against a blank screen: *This is your life without publicly-funded science.*

Similar videos that remove everything influenced by publicly-funded science can be customized to specific contexts, whether subways, buses, driving on a freeway, working a farm, inside homes, and so on, to better support public understanding of what *our lives* would be like *without* our work. QR codes displayed at the end of each video could give curious viewers access to annotated versions that identify the public funding provenance of each popular innovation - e.g., disposable diapers from the National Aeronautics and Space Administration, pacemakers from Veterans Affairs.

2.1.2 Example 2 – establishing and implementing a publicly-funded science brand strategy. Many popular public creations and resources already reference and use publicly-funded science, but attribute the work simply to individual scholars. If individual or collective scientific communities develop a ‘brand kit’ that establishes guidelines for identifying publicly-funded science in such popular resources, we can spearhead a crowdsourced effort to contact the creators of such content and ask that they use the brand kit. For example, any time a YouTube video mentions research done by a professor at a public university, or supported by public grant funding, perhaps a “made with publicly-funded science” logo can pop up briefly on the video. As another example, we could request that Wikipedia incorporate a specific tag or logo with every mention of publicly-funded science, perhaps a simplified version of the National Science Foundation’s updated branding guidelines⁵. Future work could involve pressing at state and federal levels to encourage or mandate companies to include such logos on any product that has been made possible by publicly-funded science, in order to more consistently communicate our influence to the public.

2.1.3 Example 3 - Immediate Individual Next Steps. Any scholar sufficiently advanced enough to be asked for media-based quotes or interviews can require that their contribution be accompanied by attribution to any public funding that made their expertise possible. Given the interweaving of publicly-funded science with US innovation, scholars can choose exactly which attribution to include. For example, a scientist at a private institution who does not have public funding can still explain that their work is building on work originally done by a different scholar funded by the National Institutes for Health; or a scientist at a public state university who does not have a

⁵<https://www.nsf.gov/science-matters/nsf-101-nsf-brand-identity>

funding source for their work, can still attribute their research to the public more broadly - after all, everything that emerges from a public institution is inherently publicly-supported science!

Scientists applying for funding from public, private, or internal sources should budget for the necessary expertise and work to engage the public in their science. In particular, as scientists, we are often not very expert in communicating broadly and productively with the public - after all, this is usually not part of our training! In our grant applications, we can budget for consultants that *do* have such expertise - such as science journalists, vloggers, and others who have expertise communicating science to the broader public. Finally, Science Homecoming⁶ and the McClintock letters⁷ can help you write an opinion piece for the most local paper of your childhood town - let your community know what the true cost of these cuts are.

2.2 Amplifying awareness of the ongoing damage

Recent and ongoing widespread grant cancellations (amongst other looming financial threats) are hugely impacting public research efforts and directly harming the US's standing as a leading technological powerhouse in the world. However, these cancellations are so frequent and badly communicated (and randomly un-canceled) that even scientists directly impacted have a difficult time perceiving the full scope of disaster. Instead of simply fulfilling the financial commitments the federal government previously made, taxpayer money is being wasted to defend their broken promises in court [7]. In other words, our hard-earned money—instead of going towards making life better for everyone—is going to lawyers fighting for the right to make our lives worse, more expensive, and more painful.

To highlight the true cost of these cuts, we propose increasing our engagement in public-facing communication efforts. For example, a *multi-week public teach-in series by the principal investigators of canceled grants*. The first and/or last teach-in would focus on communicating the role of public funds in innovation - that is, how much of our daily lives have been positively influenced by innovations developed through public funds. Each teach-in would be paired with an op-ed written in accessible language and a short video easily shared on social media, in order to increase the magnitude of the outreach. Another approach would be to have principal investigators of canceled grants that are around the same topic (misinformation, equity, children with cancer) craft unified content on the damage caused by the cancellations.

However, while scientists have finely honed writing and communication skills, those skills are often targeted towards peer scientists, and we need training to better communicate across audiences. For example, the teach-ins, op-eds, and short videos have different potential audiences - perhaps the teach-ins are geared towards policy-makers and politicians, while the op-eds and videos are geared towards the general public. Consequently, we recommend centralized talking points or training sessions to support presenters in sharing the work they were originally funded to do, the future impact of that work, and the damage occurring *right now* without that work. Another approach would be having a single person, skilled in broader communication, conduct interviews with scientists about their canceled grants, and 'translate' their specialized language into accessible information. For example, a *Canceled!* podcast with a knowledgeable and skilled interviewer may be able to 'translate' scientists into publicly-accessible descriptions of immediate and future impacts. This work can be partially modeled after already successful science podcasts such as *Science Fridays*⁹ or *RadioLab*¹⁰, although a drastic shift will need to be made from *celebrating science* to *describing the absence of publicly-funded science*.

⁶<https://sciencehomecoming.com/>

⁷<https://blogs.cornell.edu/asap/events-initiatives/the-mcclintock-letters/>

⁹<https://www.npr.org/podcasts/583350334/science-friday>

¹⁰<https://radiolab.org/about>

3 The Complex Social Contexts of Science

It is crucial for our scientific communities and the broader public to understand that while financial support is *necessary*, it is *insufficient* for fields on the cutting edge of innovation. To innovate is sometimes to identify scientific truths that contradict popular opinion - in other words, good science is inherently a creative and risky endeavor that requires a long term view, rather than the shorter profit-driven cycles preferred by industry. Thus far, this article has focused on how we can better communicate with the public about the value of our work, because the lack of national outrage over the brutal rescission of awarded grants illustrates just how little our contributions are recognized. However, science, like all human endeavors, does not occur in a vacuum, because science is done *by people within a social context*. If we define *doing good science* broadly as a practice of contributing sound knowledge to the world, then we must recognize that more than financial support is necessary for scientific endeavors. Rather, we need to ensure that scientists are able to focus on doing good science, and are neither seeking nor ignoring certain results simply because of anticipated approval or retribution from external non-scientific sources. It is not a large step to say, therefore, that good science benefits from scientists who are physically, socially, and intellectually safe. *Innovative* science has its own documented needs: innovation requires creativity, which benefits from intellectual experimentation and long-term (not just short-term) success [2]. In order to continue productive, innovative, *good* science, we need to *fight* against *intellectual redlining* - that is, arbitrary external forces dictating who is allowed to do what type of science, and punishing those who do not toe the line.

One way to fight intellectual redlining is to publicly highlight and emphasize canceled grants and projects in *curriculum vitae* or on research project websites, and for us to develop strategies for supporting early career and soft-money funded scholars in continuing their careers successfully. Whereas previously, canceled grants may have indicated failure by the PI to actually execute the grants, now canceled grants merely reflect the fundamental lack of scientific knowledge at higher echelons in our political system. Our early career and soft-money scholars should not have their survival and promotion dictated by powerful people who lack scientific expertise, and we should consider strategies for lessening this punishment. For example, 148 National Science Foundation CAREER grants have been reported as canceled¹¹, although the total number is unknown. These cancellations represent major disruptions to at least 148 early-career scientists who are just establishing themselves, hundreds of graduate students who will not be trained and supported through this funding, and thousands of young people who might have been inspired to learn about or pursue careers in science through the outreach activities of these now canceled grants. As one tiny step that can be taken to ensure that our academic system does not add further harm and professional stigma to these scholars, we suggest canceled grants be included on *curriculum vitae*, with a contextualizing note: *This CAREER grant was awarded and then canceled, likely due to misinformation keyword*, followed by a short paragraph about the grant and anticipated outcomes. Five years from now, as senior scholars are reviewing promotion and tenure packages, we can account for such canceled grants and their impact on scholarly trajectories.

4 Increasing Collaboration with Other Communities

The widening ripples of effect from indiscriminate terminations of congressionally approved grants and the general defunding of publicly-funded science in the USA are just starting to be felt outside of scholarly institutions, but they will worsen. As a result, we recommend that scientific communities seek partner communities that may be traditionally underrepresented/under-considered in our spaces, such as teachers, farmers, and coders. For example, librarians, firefighters, and public K-12

¹¹Grant Witness, <https://grant-witness.us>

school teachers are also struggling as the promises made by the Public Service Loan Forgiveness program are in dire jeopardy [9]. Workforce development and after-school non-profit organizations that collaborate with public K-12 schools are also losing pre-approved funding and resource partnerships [4]. Even 4-H, a beloved nation-wide program for millions of children since 1902, is on the chopping block! Farmers are suffering, as their contracts with the USDA are canceled or payments simply stop coming [3], the immigrants and citizens needed to work their farms are being summarily imprisoned even with documented legal status [10], and the administration orders the destruction of satellites that provide data that improve crop health [5]. Communities reeling from natural disasters are being repeatedly denied disaster aid to rebuild their homes and lives [1]. Transgender athletes are being denied the reality of their existence, and not even the NCAA has their backs [8]. Federal workers – many of them who served as active military on our behalf – are being fired en masse [6]. Across the board, people were promised that if they worked hard, they could have a decent life for their families – and they are being betrayed. As members of a scientific community, we have a big voice – now, we need to use it for everyone.

5 Next Steps

This article suggests only a few priorities for moving forward. We know that there are considerably more possibilities for action, as well as a constantly changing landscape that requires adept and rapid responses. Much of this is geared specifically towards what scientific communities can accomplish and how they organize. While we can individually accomplish some of these actions, our organizations need to coordinate at a broader scope, both within nations and across the world. We call on national and international scientific organizational leadership to coordinate these opportunities, to connect with other communities currently being punished alongside us, and to develop a coherent support and development system to which individual scientists can contribute.

6 Acknowledgments

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